

ASSIGNMENT #10 November 8, 2022

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SECTION 8.1

Exercise (8.1.1 — Stefano Fochesatto). For each of the following pairs of integers a and b , determine their greatest common divisor d and write d as a linear combination $ax + by$ of a and b .

a. $a = 20, b = 13$

Proof. Using the euclidean algorithm we get the following,

$$20 = 13(1) + 7$$

$$13 = 7(1) + 6$$

$$7 = 6(1) + 1$$

$$6 = 6(1) + 0$$

Which gives us $(20, 13) = 1$. By back substitution we get the following, $1 = 7 - 6 = 7 - (13 - 7) = 2(7) - 13 = 2(20 - 13) - 13 = 2(20) - 3(13)$. $\square$

c. $a = 11391, b = 5673$

Proof. Using the euclidean algorithm we get the following,

$$11391 = 5673(2) + 45$$

$$5673 = 45(126) + 3$$

$$45 = 3(15) + 0$$

Which gives us $(11391, 5673) = 3$. By back substitution we get the following, $3 = 5673 - 45(126) = 5673 - (11391 - (2)5673)(126) = 253(5673) - 126(11391)$ $\square$

e. $a = 91442056588823$, $b = 779086434385541$

Proof. Using the euclidean algorithm we get the following,

$$779086434385541 = 91442056588823(8) + 47549981674957$$

$$91442056588823 = 47549981674957(1) + 43892074913866$$

$$47549981674957 = 43892074913866(1) + 3657906761091$$

$$43892074913866 = 3657906761091(11) + 3655100541865$$

$$3657906761091 = 3655100541865(1) + 2806219226$$

$$3655100541865 = 2806219226(1302) + 1403109613$$

$$2806219226 = 1403109613(2) + 0$$

Which gives us $(91442056588823, 779086434385541) = 1403109613$. By back substitution we get, $1403109613 = 277522(91442056588823) - 32573(779086434385541)$ $\square$

Exercise (8.1.7 — Stefano Fochesatto). Find the generator for the ideal $(85, 1 + 13i)$ in $\mathbb{Z}[i]$, i.e. the greatest common divisor for 85 and $1 + 13i$, by the Euclidean Algorithm. Do the same for the ideal $(47 - 13i, 53 + 56i)$.

Solution. As outlined in Example 3 of Section 8.1 we know that The Gaussian integers are a Euclidean Domain under the norm $N(a + bi) = a^2 + b^2$. Recall the division algorithm for The Gaussian integers under norm N , for all $\alpha, \beta \in \mathbb{Z}[i]$ there exists some $(p + qi), \gamma \in \mathbb{Z}[i]$ such that $\alpha = (p + qi)\beta + \gamma$ where $N(\gamma) \leq \frac{1}{2}N(\beta)$. We begin the Euclidean algorithm, note that

$$\frac{85}{1 + 13i} = \frac{85(1 - 13i)}{(1 + 13i)(1 - 13i)} = \frac{85 - (85)13i}{170} = \frac{1}{2} - \frac{13}{2}i.$$

Choosing a close $(p + qi) \in \mathbb{Z}[i]$ we get $(p + qi) = 1 - 7i$. Finding the product $(1 + 13i)(1 - 7i) = 92 + 6i$. Solving for $\gamma = 85 - (92 + 6i) = -7 - 6i$. Double checking norms we get, $N(-7 - 6i) = 7^2 + 6^2 = 85 \leq 85 = \frac{1}{2}170N(1 + 13i)$. So our first step looks like,

$$85 = (1 - 7i)(1 + 13i) + (-7 - 6i).$$

Computing the next step we first check the quotient,

$$\frac{1 + 13i}{-7 - 6i} = \frac{(1 + 13i)(-7 + 6i)}{(-7 - 6i)(-7 + 6i)} = \frac{-85 - 85i}{85} = -1 - i.$$

Which gives us our final step,

$$1 + 13i = (-7 - 6i)(-1 - i) + 0.$$

Therefore the greatest common divisor for 85 and $1 + 13i$ is $(-7 - 6i)$.

Applying the Euclidean algorithm for $47 - 13i$ and $53 + 56i$, first we compute the following quotient,

$$\frac{53 + 56i}{47 - 13i} = \frac{(53 + 56i)(47 + 13i)}{(47 - 13i)(47 + 13i)} = \frac{1763 + 3321i}{2378} = \frac{43}{58} + \frac{81}{58}i.$$

Choosing a close $(p + qi) \in \mathbb{Z}[i]$ we get $(p + qi) = 1 + i$. Finding the product $(47 - 13i)(1 + i) = 60 + 34i$. Solving for $\gamma = (53 + 56i) - (60 + 34i) = -7 + 22i$. Double checking norm we get, $N(-7 + 22i) = 533 \leq 1189 = \frac{2378}{2} = \frac{1}{2}N(47 - 13i)$. So our first step looks like,

$$53 + 56i = (1 + i)(47 - 13i) + (-7 + 22i).$$

Computing the next step we first check the quotient,

$$\frac{47 - 13i}{-7 + 22i} = \frac{(47 - 13i)(-7 - 22i)}{(-7 + 22i)(-7 - 22i)} = \frac{-615 - 943i}{533} = -\frac{15}{13} - \frac{23}{13}i.$$

Choosing a close $(p + qi) \in \mathbb{Z}[i]$ we get $(p + qi) = -1 - 2i$. Finding the product $(-7 + 22i)(-1 - 2i) = 51 - 8i$. Computing $\gamma = (47 - 13i) - (51 - 8i) = -4 - 5i$. So our next step

is

$$47 - 13i = (-1 - 2i)(-7 + 22i) + (-4 - 5i).$$

For the next step we compute the following quotient,

$$\frac{-7 + 22i}{-4 - 5i} = \frac{(-7 + 22i)(-4 + 5i)}{(-4 - 5i)(-4 + 5i)} = \frac{-82 - 123i}{41} = -2 - 3i.$$

Which gives us our final step,

$$-7 + 22i = (-2 - 3i)(-4 - 5i) + 0.$$

Therefore greatest common divisor for $47 - 13i$ and $53 + 56i$ is $(-4 - 5i)$.

□

SECTION 8.2

Exercise (8.2.1 — Stefano Fochesatto). Prove that in an Principal Ideal Domain R , two ideals (a) and (b) are comaximal if and only if a greatest common divisor of a and b is 1.

Proof. Suppose R is a Principal Ideal Domain and two ideal (a) and (b) are comaximal. By the definition of comaximal we know that $(a) + (b) = R$. By Proposition 6 we know that if d is a generator for the principal ideal generated by a and b then $d = \gcd(a, b)$. Therefore $(a), (b) \subseteq (d)$ and by (ii) of Proposition 6, $(a) + (b) \subseteq (d)$. Since (a) and (b) are comaximal we know that $(a) + (b) = R$ therefore $(d) = R$. Thus $d = 1$ and therefore $\gcd(a, b) = 1$.

Suppose R is a Principal Ideal Domain with elements a and b such that $\gcd(a, b) = 1$. Consider (a) and (b) by (ii) of Proposition 6 we know that $(1) = (a)x + (b)y$ where $x, y \in R$. Thus (a) and (b) are comaximal.

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